Republic of the Philippines
**CAVITE STATE UNIVERSITY**
**Don Severino de las Alas Campus**
Indang, Cavite

**COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY**
**Department of Information Technology**

Name: _____ Student No.: _____
Course / Yr. / Sec.: _____ Date: _____

**Portfolio #1 – Cybersecurity Cube Analysis – A Collaborative Learning Assessment**

## Objectives

- To collaboratively apply the principles of the CIA Triad to a realistic software development scenario.
- To map the states of data and implement appropriate countermeasures using the McCumber Cybersecurity Cube.
- To foster teamwork and critical thinking in designing secure system architectures.

## Background / Scenario

*This is a Collaborative Learning Assessment. You must work in groups of 3 to 5 members.*

Your group acts as the core security architecture team for **"CampusPay,"** a newly proposed e-wallet application designed exclusively for Cavite State University students. The app allows students to pay for cafeteria meals, library printouts, and university merchandise using their smartphones. The app links directly to the university's central database for student verification and uses a third-party payment gateway for fund top-ups.

Before the development phase begins, your team must present a foundational security plan based on the **McCumber Cybersecurity Cube**.

## Instructions

As a group, discuss and answer the following sections. Assign a scribe to document your finalized answers.

### Part 1: The CIA Triad Application

Describe how your team will ensure the following principles specifically for the *CampusPay* application:

- **Confidentiality:** How will you ensure that a student's e-wallet balance and transaction history remain hidden from other students on the same campus Wi-Fi?
- **Integrity:** What mechanism will ensure that if a student buys a ₱50 meal, exactly ₱50 is deducted from their account and exactly ₱50 is credited to the cafeteria, without any modification in transit?
- **Availability:** During midterm week, thousands of students will use the app simultaneously. How will your team design the system's infrastructure to handle this load without crashing?

### Part 2: States of Data Mapping

Identify where the data exists in the three states during a standard transaction, and specify *one* technical vulnerability for each state:

- **Data at Rest:** (Where is the data stored, and what is the risk?)
- **Data in Transit:** (How is the data moving, and what is the risk?)
- **Data in Process:** (Where is the calculation happening, and what is the risk?)

**Part 3: Designing Countermeasures**
Based on your answers in Part 2, propose one specific countermeasure for each dimension of the Z-axis:

- **Technology:** Propose one hardware or software control to protect the data in *Transit*.
- **Policy:** Draft one specific rule (1-2 sentences) regarding employee access to the *Data at Rest*.
- **People:** Propose one training or awareness initiative for the students using the app to protect their accounts.

**Reflection (Individual)**

*Note: Each member must write a short, 3-4 sentence reflection.*

1. Based on your group discussion, which aspect of the Cybersecurity Cube do you believe is the most difficult for software developers to implement correctly, and why?

   Name1: Reflection…

   Name2: Reflection…

**Submission Format**

SUBMIT YOUR ANSWERS in a single PDF document and remove all headers and rubric in your submission. Filename: **Portfolio1_GroupName_Section.pdf** (e.g., Portfolio1_Group1_BSCS-3-1.pdf). Include all members' names and their individual reflections at the end of the document.

**ONLY 1 GROUP MEMBER WILL SUBMIT PER GROUP**

Rubric

| Criterion | Excellent (25 Points) | Good (16-20 Points) | Needs Improvement (11-15 Points) | Poor (0-10 Points) | Points |
|---|---|---|---|---|---|
| 1. Following Instructions & Formatting | All parts of the activity (Parts 1-3 and Individual Reflections) are fully completed. The PDF is correctly named and perfectly structured with all required sections clearly labeled. | All major parts are completed, but there is a minor omission (e.g., one member missed a reflection) or a slight deviation from the required format. | A significant portion of the submission is incomplete, or the document is poorly formatted making it difficult to read. | The activity is not submitted, or it completely disregards the instructions and format. | /25 |
| 2. AI Originality & Collaborative Effort | The submission demonstrates clear, original group thought. Explanations are contextualized perfectly to the "CampusPay" scenario. Individual reflections show genuine insight. No signs of AI generation. | The work appears to be the group's own and connects to the scenario, but the analysis is somewhat generic. The language is simple and lacks boilerplate phrasing. | The submission is vague, overly formal, or contains generic phrasing that raises concerns about originality and the potential use of AI tools for generation. | The submission is clearly generated by an AI tool with little to no modification, is plagiarized, or shows no evidence of group collaboration. | /25 |
| 3. Application of the CIA Triad & Data States | Deep understanding of C-I-A and data states; examples are technically sound, highly relevant to a mobile payment app, and demonstrate CS-level knowledge. | Solid understanding; mostly accurate but lacks minor technical detail or relies on overly basic examples. | Basic understanding; confuses some principles of the triad or misidentifies the states of data. | Major inaccuracies or failure to define security principles correctly within the scenario. | /25 |
| 4. Countermeasure Design | Proposes highly realistic, effective, and specific solutions for Technology, Policy, and People that directly mitigate the identified risks. | Proposes logical countermeasures, but they are generic (e.g., just saying "use a firewall" without context). | Solutions provided do not directly address the risks identified in Part 2, or they confuse Policy with Technology. | No countermeasures proposed, or they are non-sensical and ineffective. | /25 |
| TOTAL SCORE | | | | | /100 |